

up gold but in the intellectual and physical strength of its people.”

BRIGHT SPARK

He was among the finest teachers of science in the country. Not only did he do great science himself but he also made others do great science.

■ By M.G.K. Menon

When the Republic Day Awards were instituted in 1954, C.V. Raman was bestowed with the Bharat Ratna, the very first to be given it, indicative of his standing among all the great personalities of the country of that time. His was already a household name by then. When it was decided to observe a National Science Day, the date chosen was February 28, when the Raman Effect was discovered. Raman is the only scientist whose work, entirely carried out in India, got him the Nobel Prize in physics in 1930.

I first met him in Jodhpur when my father had invited him for a meal. I was then studying for my B.Sc. (1944-46). For a person in the mid-teens, it was quite an experience to meet someone like him. He was exuberant and completely focused. He thumped the table and said, “the greatest thing for you to do in life would be to do science”. He insisted that I should go on to do research in physics—a subject I was good at. Anyone who came in contact with him would be infected by his ideas, his sense of excitement and enthusiasm.

Many years later, in 1967, at the annual meeting of the Indian Academy of Sciences in Madras, just before his presidential address, Raman turned around and said, “Menon, Ramanna, you see this great hall packed with people who have all come to hear me. Can you fill this hall with such an audience?” He was proud of the fact that for a science lec-

ture there could be such a large audience. But he also displayed humility when he spoke about others whom he admired, such as Newton, Einstein, Rutherford, Rayleigh and Boltzmann. He was in the line of these great classical scientists—a dwindling band. The richness of his life and discoveries, and his outstanding qualities covered the canvas of Indian science in dominating fashion for six decades.

Raman was a self-made man; he was a precocious student, and did extremely well in his studies throughout. He had a flair for science and also an unusual appreciation of English literature. As a result, he was one of the most brilliant lecturers on science one could listen to. I recall the annual session of the Indian Academy of Sciences in Ahmedabad in 1968, to celebrate his 80th birthday, when he stood under a tree talking to a huge crowd of children. There was no blackboard and, indeed, no one could have seen anything written from the distance where the children were sitting. There was only one microphone to reach out to the crowd. He looked up to the sky above and asked them, “why is the sky blue in colour?” He then spent the next hour expounding on that and held them all spellbound.

He was fascinated by colour, form and geometry. Often, when giving lectures, he would look at the bright saris of the women in the audience, and remark on them in his inimitable jocular style. His admiration for beauty and the understanding of the scientific essence comes out in what he said: “The beautiful silky hair which is the

crowning ornament of a woman's head owes its geometrical character of a long continuous fibre to the structure of a certain type of protein molecule. The glory of human hair thus stands revealed as a special effort in protein chemistry made by nature.”

Unbounded curiosity, single-minded devotion and commitment, deep physical insight, supreme self-confidence and courage of his convictions were the hallmarks of his character. He appeared for a competitive examination when he was 18 and got selected for the finance department. On his way to work by tram in Calcutta, he saw a signboard which read: “Indian Association for the Cultivation of Science”. He started working in the laboratory of the institution set up by Mahendra Lal Sircar in 1876. This resulted in work of very high quality in the fields of acoustics and optics. On an offer by Sir Ashutosh Mukherjee of the University of Calcutta, Raman left government service to become the first Palit professor of physics. In Calcutta, he collected a very large number of younger scientists who came to work with him. After retirement from the Indian Institute of Science, Bangalore, he set up the Raman Research Institute, where he spent the last years of his life. Not only did he do great science himself but he also made others do great science. It is difficult to think of another who had so many qualities.

The author is a physicist and former Union minister of state for science and technology